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Morphological diversity and disparity in trilobite cephala and the evolution of trilobite enrolment throughout the Palaeozoic

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Morphology permits the extracting of information to study patterns of disparity and diversity of a particular group of animals through time. Enrolment is a characteristic behaviour of trilobites, having been first recorded in the Cambrian and continued until their demise at the end of Permian mass extinction. Morphological trends related to enrolment strategies have been documented previously but are yet to have been analysed in a morphospace framework. Here we analyse the cephalic shape of a wide range of trilobite taxa using geometric morphometrics. These methodologies were implemented to explore whether the relationship between the shape of the head and the evolution of different enrolment styles correspond with an increase in morphospace or disparity during the Palaeozoic. Our results show distinct patterns of diversity and disparity in the functional morphology of trilobites during the Palaeozoic. Furthermore, these analyses show that enrolment types in trilobites without interlocking devices (e.g. cylindrical) permit more flexibility in the head shape, promoting higher rates of disparity, possibly leading to an enhanced ability to occupy a broad range of ecological niches. On the contrary, lower disparity cephalic shape constrains the way to enclose the body and consequently places limitations on the ability to adapt to new niches. Our results indicate that the higher rates of morpho-functional diversity and disparity coincide with the increase of morphospace in radiations in the early Palaeozoic (Cambrian and Ordovician) but also immediately after the Carboniferous-Permian biodiversification event. Low disparity rates constrain the success of trilobites with a specific enrolment type. Finally, the results show that sphaeroidal enrolment style was the most successful type that evolved independently in all trilobite orders. □ *Arthropods, ecology, ecomorphology, evolution, functional morphology.*

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Enrolment is a common behaviour in many animals. Typically viewed as a defensive response, enrolment is first recorded in animals in the Cambrian and continues through the Phanerozoic to modern animals. Today, vertebrates such as hedgehogs (Heatley 2009), pangolins (Tenaza 1975) and armadillos (Superina & Loughry 2012), and invertebrates, such as isopods (Béjaoui *et al.* 2019), shrimp larvae (Haug & Haug 2014) and millipedes (Golovatch & VandenSpiegel 2014; Tuf *et al.* 2016), show this defensive behaviour. Trilobites were the first animals to implement this adaptative behaviour (Ortega-Hernández *et al.* 2013), presenting specialized morphological features (coaptative structures) in their bodies that allowed for enrolment to be executed (Chatterton & Campbell 1993; Clarkson & Whittington 1997; Esteve *et al.* 2011). Due to the extensive documentation of enrolment strategies from Cambrian Series 2 (Palmer 1958; Ortega-Hernández *et al.* 2013) through to the Permian (Lerosey-Aubril & Anfiolini 2009; Fortey & Heward 2014), enrolment provides important morpho-functional information to track evolutionary trends (see Esteve *et al.* 2011; Esteve & Yuan 2017). During the Palaeozoic, numerous types of trilobite enrolment have been recognized, clearly defined and classified (Bergström 1973; Chatterton & Campbell 1993; Esteve 2013).

Morphological variations and have been observed in trilobites previously, showing increased diversity and disparity that correlate with the main biodiversification events during the Palaeozoic (e.g. Foote 1991, 1993; Lerosey-Aubril & Feist 2012; Adrain 2013). But the morphological variation and disparity in trilobites have never been analysed under a morpho-functional framework. Here, we described the morphology variations and disparity in the head

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of four hundred ($n = 400$) trilobite taxa (including agnostids) throughout the Palaeozoic, within a morpho-functional framework (i.e. enrolment). By analysing the head outline in trilobites, we aim to explore whether the morpho-functional novelties and evolution of different enrolment styles coincide with an increase in morphospace during radiation events and/or variations in the morphological disparity of this group of arthropods.

Materials and methods

Data set construction

Four hundred ($n = 400$) cephalic outlines have been used for our study. Enrolled trilobites have been published for over 250 trilobite taxa, from strata ranging from the early Cambrian (~516 Ma) to Permian in age (~251 Ma). Enrolment of trilobites without direct evidence (i.e. preservation of enrolled specimens) has been inferred by morphological comparison with taxa where direct evidence is known. These cephalic outlines include representatives of all orders and span a broad range of environmental settings. Orders are based on Fortey (1997), but we bring the Suborder Harpina (*sensu* Fortey 1997) out from the Order Ptychopariida given the reasonable differences of this group of trilobites and following Adrain (2011). Given that many authors have argued that the Order Ptychopariida is problematic and maybe a wastebasket taxon (see Fortey 1997, Adrain 2011), we used here the term 'ptychopariid'. Our dataset is based on data that we have personally acquired, some of which has already been published (Esteve *et al.* 2011; Esteve 2013; Esteve & Yuan 2017) and supplemented by published records that include all papers providing relevant information known to us published to December 2020 (see Appendix S1). In addition, our dataset has been further updated using information from the Palaeobiology Database (PBDB, December 2020).

The images were edited using Adobe Illustrator 2021 to produce the outline models, used for the semi-landmark analysis. We included the free cheeks in our analysis as they play an important role during enrolment and different enrolment styles are closely related to the shape of the free cheeks (see Bergström 1973; Esteve *et al.* 2010, 2012). The presence of doublures in the free cheeks is also important as they bear different structures such as vincular furrows, notches or girder where pleural tips fit or rest, depending on the enrolment style (see Bergström 1973; Chatterton & Campbell 1993; Stitt 1983; Bruton & Haas 1997; Esteve 2013; Esteve *et al.* 2010, 2012, 2017, 2018). Here, we use the enrolment styles according to Esteve (2013) that are briefly explained below (Fig. 1):

- *Cylindrical type*. Enrolment where the pleural spines fail to meet laterally, typical in some redlichiine, olenelloids and odontopleurids (see Bergström 1973; Hammann & Rábano 1987; Whittington 1989; Esteve & Yuan 2017).
- *Sphaeroidal type*. Enrolment where the pygidium is resting beneath the cephalon or fitting into a vincular furrow in the cephalic doublure, typical of many different trilobite taxa such as 'ptychopariids', phacopids, proetids (see Stitt 1983; Chatterton & Campbell 1993; Whittington 1993; Bruton & Haas 1997; Esteve *et al.* 2010, 2017; Esteve 2013; Yuan *et al.* 2014) but also exclusively in agnostids (Bruton & Nakrem 2005; Esteve & Zamora 2014).
- *Spiral type*. Enrolment in which the pygidium is hidden beneath the ventral side of the cephalon, typical in many 'ptychopariids' (see Esteve 2013; Esteve *et al.* 2017).
- *Unenrolled type*. Type of enrolment similar to spiral but it is partly unrolled, so the proximal part of the pygidium is visible even in the full enrolment specimen, typical in many calymenids (see Hammann 1983, 1985; Esteve *et al.* 2018).
- *Inverted type*. Enrolment where the pygidial spines or the rear part of the pygidium reach the dorsal side of the cephalon, typical in some phacopids (see Clarkson & Henry 1973; Esteve *et al.* 2018).
- *Discoidal type*. Enrolment where the rear part of the pygidium is fitting into a furrow under a narrow geniculated groove below a girder subparallel to the anterior cephalic border, typical in harpids and some asaphids (see Stitt 1983; Chatterton & Campbell 1993; Esteve *et al.* 2012).

Geometric morphometrics and statistical analysis

Geometric morphometric methods were performed on the cephalon of each individual in the dataset. Markers or landmarks were placed as curves to completely envelop the shape of the cephalon, these were processed using tpsDig v.2.32. For creating curves, 90 semi-landmarks were employed for each sample, semi-landmarks were allocating from the middle of the anterior border (axial) following clockwise along the whole border (see Appendix S1). In order to stan-

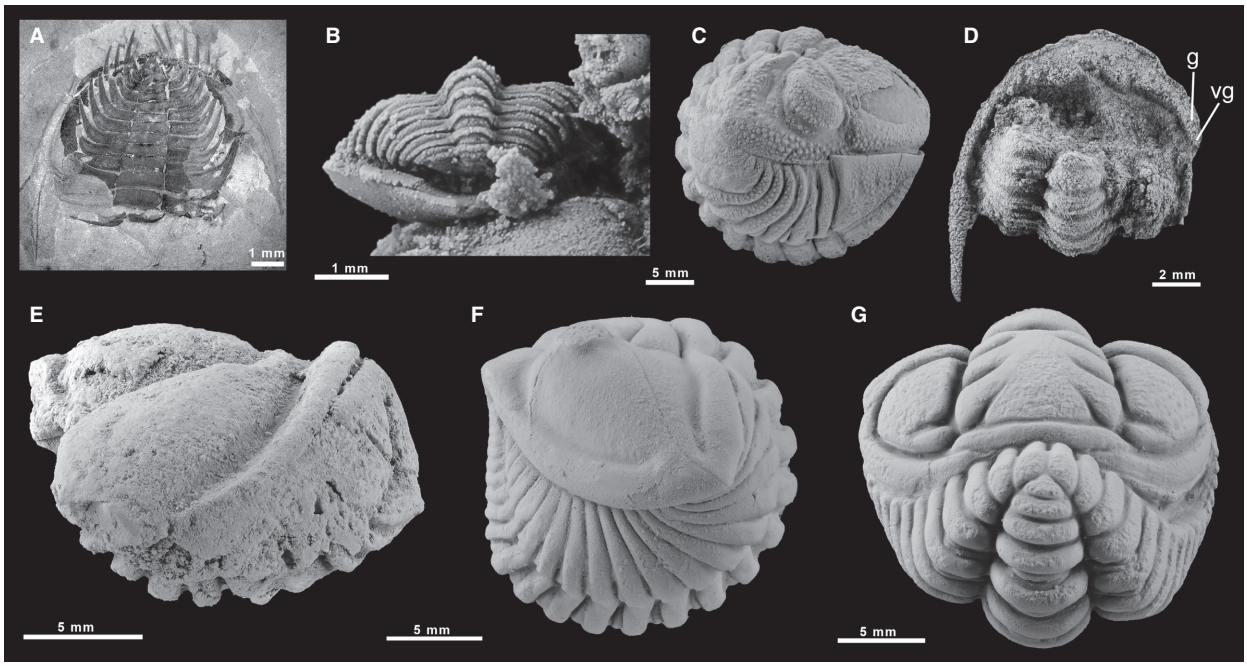


Fig. 1. Trilobite enrolment types. A, cylindrical enrolment type in the olenellid *Olenellus chiefensis* (DMNH 16080) from the early Cambrian (upper Dyeran) of the Pioche Formation, Nevada, USA. B, spiral enrolment type in *Bailiaspis glabrata* (MPZ 2012/57) from the Guzhangian siliceous concretions of the Valtorres Formation, Purujosa, northeastern Spain. C, sphaeroidal enrolment type in *Phacops (Chotecops) zizensis* (MPZ 20013/50) from the Eifelian of Tafilalt, Morocco; the pygidium is fitting into the cephalic vincular furrow. D, discoidal enrolment type in *Schopfaspis graciai* (MPZ 2009/350) from the Drumian of the Murero Formation of the Purujosa trilobite assemblage, northeastern Spain, showing the vincular groove (vg) and the girder (g). E, sphaeroidal enrolment type in *Parabailella langudocensis* (C-12) from the hill slope of Vinice near Jince of the Barrandian area, Czech Republic, the pygidium is resting beneath the cephalon but not fitting into a furrow as in C. F, unenrolled enrolment type in *Neseuretus avus* (NEIV-19) from the Navas de Estena Formation of the Valderuelo site, Mounts of Toledo area of Central Spain. Early Oretanian (=middle Darriwilian) age. G, inverted enrolment type in *Placoparia (P.) cambriensis* (VPA-81) from the Navas de Estena Formation at the Acebrón stream fossil locality (south of Las Ventas con Peña Aguilera, province of Toledo), Mounts of Toledo area of Central Spain, early Oretanian (=middle Darriwilian).

dardize the size, orientation and position of each specimen (and thus the alignment of each landmark), a Procrustes superimposition of data was applied before analytical processing (the resulting coordinates are named, after fitting, Procrustes coordinates). Coordinates of semi-landmarks were calculated from the outline data in SemiLand8 (Sheets, 2014) using the minimized Procrustes distance method to optimize their location along the outline (see for more details Sheets *et al.* 2004; Webster & Zelditch 2011; Webster 2011; Álvaro *et al.* 2018). The resulting Procrustes aligned landmark coordinate data of the extant data were exported and statistically processed using Python 3.6 (see Appendix S1). The libraries and packages used for data analysis were numpy, pandas, scipy, sklearn and stats models to generate basic statistics, the principal components analyses (Pedregosa *et al.* 2011) and sum of variances, the last of which was calculated using the three principal components scores implying the full morphospace (Ostrowski & Menyhárt 2020; Schaeffer *et al.* 2020). Furthermore, matplotlib, mpl_toolkits and seaborn provided data visualization.

Results

Dataset analyses

The analysis of our sample ($n = 400$) shows a higher abundance of trilobite taxa in the Cambrian (ca. 29.5%) and Ordovician (ca. 35%) and lower at the Silurian (ca. 6%), Carboniferous (ca. 6%) and Permian (ca. 8%) with a small increase in the Devonian (ca. 16%; Fig. 3). Cambrian and Ordovician taxa show the highest diversity of enrolment styles (Fig. 2 A). During the Cambrian, up to five different enrolment styles were established: sphaeroidal (ca. 34%), unrolled (ca. 27%), spiral (ca. 27%), cylindrical (ca. 11%) and discoidal (ca. 1%). Throughout the Ordovician up to 6 different enrolment types have been found: sphaeroidal (ca. 50%), spiral (ca. 19%), unrolled (ca. 11%), discoidal (ca. 15%), inverted (ca. 4%) and cylindrical (ca. 1%). Trilobites in the Silurian display four kinds of enrolment, sphaeroidal (ca. 65%), discoidal (ca. 17%), cylindrical (ca. 9%) and unrolled (ca. 9%) and in the Devonian up to five different styles have been found: sphaeroidal (ca.

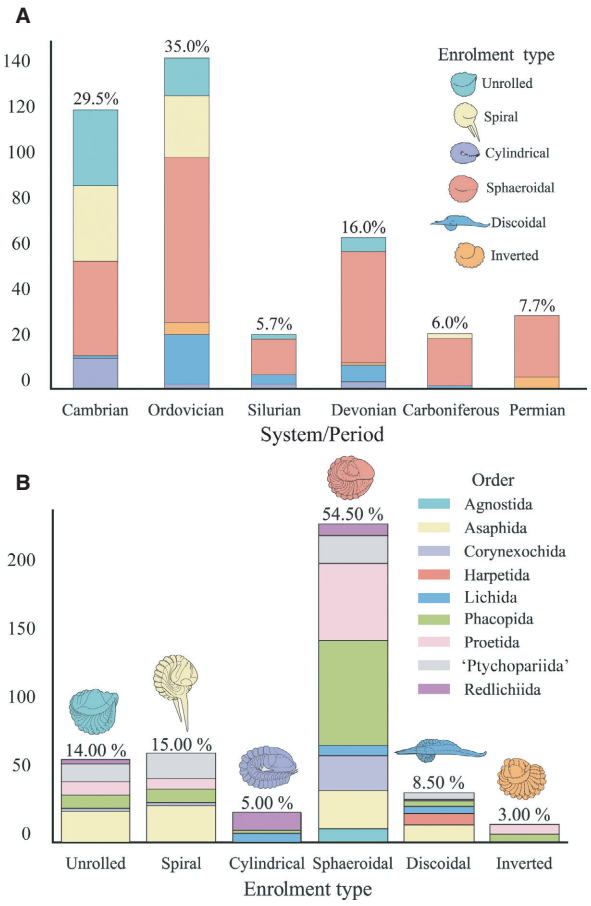


Fig. 2. Enrolment type distribution throughout the Palaeozoic (A) and among Trilobite Orders (B).

73%), discoidal (ca. 11%), unrolled (ca. 9%), cylindrical (ca. 4.7%) and inverted (ca. 1.6%). Three different enrolment types occur in the Carboniferous: sphaeroidal (ca. 83.3%), spiral (ca. 9.3%) and discoidal (ca. 7.4%), and only two different types have been seen in the Permian: sphaeroidal (ca. 84%), and inverted (ca. 16%).

Figure 2B shows the relationship between the trilobite orders and enrolment types. Our results show that the sphaeroidal enrolment type was employed by 54.5% ($n = 218$) of taxa in our dataset with the remaining enrolment types showing relatively low percentages; spiral (ca. 15%), unrolled (ca. 14%), discoidal (ca. 8.5%), cylindrical (ca. 5%) and inverted (ca. 3%). Focussing on the different trilobite orders, redlichiids enrolled using cylindrical (ca. 50%), sphaeroidal (ca. 33%), unrolled (ca. 12.5%) and spiral (ca. 4%). 'Ptychopariids' enrolled using sphaeroidal (ca. 36%), spiral (ca. 32%), unrolled (ca. 23%) and discoidal (ca. 7.5%). Proetids enrolled predominantly using sphaeroidal (ca. 70%), but also spiral (ca. 9%), unrolled (ca. 12%) and inverted (ca. 8%). Phacopids also predominantly enrolled using sphaeroidal (ca. 71%), with lower percentages of spiral (ca. 9%), unrolled (ca. 9%), inverted (ca. 5%), discoidal (ca. 4%) and cylindrical (ca. 2%). Odontopleurids enrolled using discoidal (ca. 28%), cylindrical (ca. 33%) and sphaeroidal (ca. 40%). Harpids enrolled using discoidal (100%). Corynexochids enrolled using spiral (ca. 7%), unrolled (ca. 7%) and sphaeroidal (ca. 86%). Asaphids enrolled using spiral (ca. 29%), unrolled (ca. 25%), sphaeroidal (ca. 31%) and discoidal (ca. 14.5%). Finally, agnostids enrolled using only the sphaeroidal enrolment strategy.

Morphological analysis

The three first principal axis account ca. 93% of the cephalic shape variance (Fig. 3). Variation in morphospace of trilobites with several enrolment styles shows different trends throughout the Palaeozoic. Morphospace in the Palaeozoic is clearly dominated by the sphaeroidal enrolment type. Trilobites with this enrolment style occupy a large portion of morphospace in the Cambrian and the amount of morphospace decreases until the Silurian. After the Silurian, morphospace occupied by the sphaeroidal enrolment type incrementally increases until the Permian. The unrolled enrolment type also occupies a large portion of morphospace in the Cambrian. However, it decreases during the Ordovician and Silurian, and although the unrolled type increases slightly in the Devonian, it disappears entirely in post-Devonian strata. The spiral enrolment type also occupies a large area of morphospace in the Cambrian, decreasing in the Ordovician and keeping approximately the same area in the Silurian, and trilobites used in this style are not recorded after the Silurian. Morphospace of the cylindrical enrolment style is relatively small and overlaps with the morphospace shown by the sphaeroidal enrolment type. It occupies only a small morphospace in the Ordovician and Silurian and there are only a few records of its kind existing in the Devonian. The discoidal enrolment style appears in the Cambrian and although during the Ordovician its morphospace overlaps with that of the sphaeroidal, unrolled and spiral morphospace; it continues to show clear differentiation in morphospace. The discoidal enrolment style becomes better established in the Silurian, with many specimens showing a shape distinct from the mean shape of this style. During the Devonian, the morphospace of the trilobites enrolments discoidal style overlaps with the other styles; however, we can identify some samples away from the trend. Finally, the inverted style appears in the Ordovician and occupies a small amount of morphospace during this

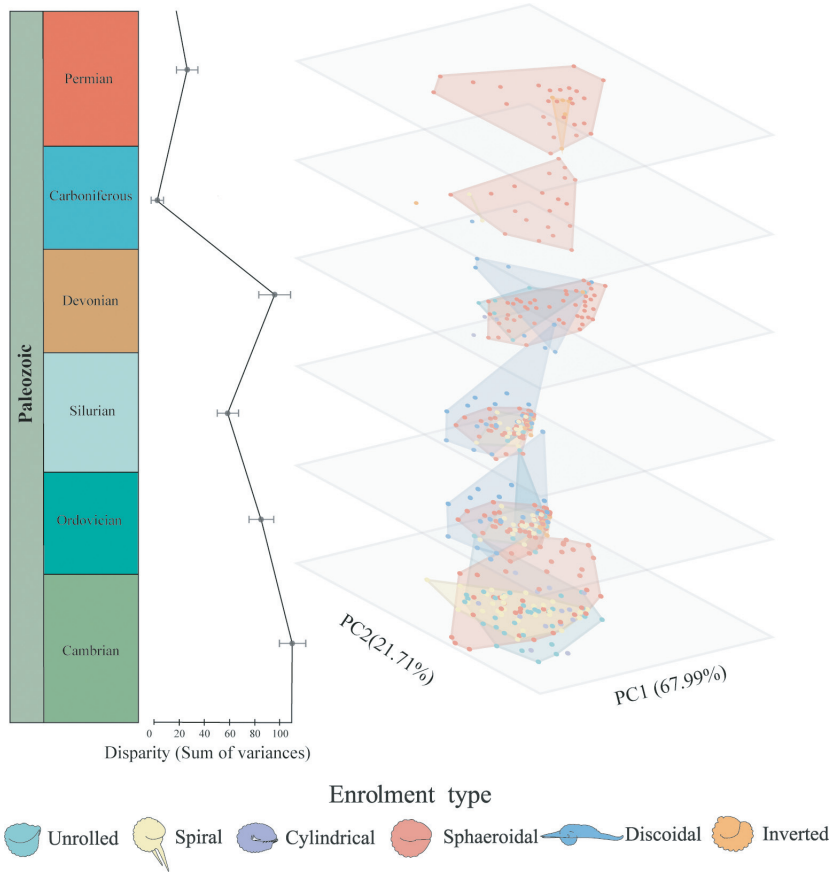


Fig. 3. Patterns of morphological disparity (sum of variances, SOV) and morphospace occupation (principal components analysis, PCA) through Palaeozoic Era, the six enrolment types are labelled.

period and the Silurian. This style is relatively reduced during the middle Palaeozoic but increases noticeably during the Permian.

Disparity analysis

Overall, the morphological disparity analysis of the cephalic outline shows a higher degree of disparity during the Cambrian. This disparity decreases during the Ordovician and Silurian rebounds in the Devonian followed by a lower morphological disparity during the Carboniferous with a slight increase during the Permian (Fig. 3). With regard to the morphological disparity of the cephalic shape in trilobites with different enrolment types throughout the Palaeozoic, we can observe that those trilobites possessing the discoidal enrolment type have the highest disparity (Fig. 4A). The morphological disparity of the cephalon in trilobites with unrolled, spiral, cylindrical and sphaeroidal ranges from SOV = 85 to SOV = 100 (Fig. 4A). The morphological disparity in the cephalon of trilobites with the inverted enrolment type is the lowest in our dataset (Fig. 4A). However, the evolution of the cephalic outline disparity in trilobites sharing the same enrolment type is not uniform (Fig. 4B–F). During the Cambrian, trilobites displaying the sphaeroidal enrolment type show the highest disparity (SOV = 165), while trilobites with unrolled, spiral, and cylindrical show similar lower values (Fig. 4B). During the Ordovician, the morphological disparity in the outline of the head of trilobites that enrol via the cylindrical enrolment technique reaches high values in our analysis (Fig. 4C). This trend of high disparity in the head of trilobites with the cylindrical enrolment type continues through to the Silurian (Fig. 4D). During the Ordovician and Silurian, the morphological disparity of trilobites with sphaeroidal and unrolled enrolment types show similar values during these two periods (Fig. 4C,D). During the Devonian, the head disparity of trilobites possessing sphaeroidal and unrolled types are similar to those values seen in the Silurian, whereas the cylindrical enrolment type shows very low values (Fig. 4E). Discoidal enrolment was very poorly developed during the Cambrian with only one trilobite documented exhibiting such a method. But trilobites with this enrolment type show a high morphological disparity from the Ordovician

until the Devonian (Fig. 4C–E) although this dramatically reduces in the Carboniferous (Fig. 4F). Morphological disparity of the cephalic shape in trilobites with the spiral enrolment type decreases with respect to those observed in the Cambrian (Fig. 4C). Spiral enrolment type is not recorded in our sample again until the Carboniferous, although with a very low morphological disparity. (Fig. 4F).

Discussion and conclusions

The results of our analysis show a greater diversity in the number of analysed specimens during the Cambrian and Ordovician. These two periods also show the highest diversity of enrolment types, and these results agree with those reported by Esteve *et al.* (2011). Stratigraphically speaking, early trilobites such as the redlichiids and 'ptychopariids' show the highest diversity in the way that they can enclose their body. Possibly, the presence of empty niches and the possibility to explore different ways to protect themselves can explain the reason behind this high diversity of enrolment types during the Cambrian and Ordovician. Laibl *et al.* (2016) showed how paradoxidid trilobites used slightly different enrolment types. These authors argued that differences in the environment and high independent evolutionary rates can explain such differentiation in paradoxid enrolment style. On the other hand, our results do not show a clear differentiation between the Cambrian and the Ordovician in terms of the number of enrolment styles. However, all the five enrolment styles which appeared in the Cambrian changed dramatically with the increase in the number of interlocking devices during the Ordovician (Esteve *et al.* 2011) and only one style (i.e. inverted) appeared in the Ordovician. However, the discoidal enrolment type only appears in one 'ptychopariid' trilobite in the Cambrian and very likely this represents a convergence adaptation. Phylogenetic connection between this Cambrian taxa and Ordovician trilobites is not clear but likely they represent unrelated species; therefore, we could say that in fact two enrolment types appear in the Ordovician. Esteve *et al.* (2018) argued that the inverted enrolment type is the only style rooted in the Ordovician and other styles like the unrolled technique are based on small modifications of previous styles that were rooted in the Cambrian. However, our analysis here shows that the unrolled type was also very common during the Cambrian and only small modifications were recorded in Ordovician trilobites (see Hammann 1985; Esteve *et al.* 2018). Esteve *et al.* (2011) additionally showed that the number of coaptative devices increased during the Ordovician and remained stable during the rest of the Palaeozoic. Here, we show that such an increase in the number of coaptative devices is accompanied by an increase in the morphological disparity of the trilobite head in some enrolment styles (cylindrical and unrolled), together with the evolution of two enrolment types (discoidal and inverted). Esteve *et al.* (2011) did not show trilobites with the cylindrical enrolment type in the Ordovician and Silurian, rather these types were included within the sphaeroidal category. This work has enlarged the dataset in an attempt to fix this problem. Many of those trilobites displaying the cylindrical enrolment type in the Ordovician and Silurian belong to the order Lichida, after Fortey (1997) [or Odontopleurida, after Harrington and Moore (1959); see Whittington (2002) for details of systematic differences between these two orders] and show high morphological disparity (Fig. 4C,D). These trilobites lack interlocking devices in the head and bear many modifications in the pleural segments and spines (see Esteve & Yuan 2017). Esteve & Yuan (2017) argued that the lack of coaptative devices of odontopleurids in combination with a very conservative body plan somehow constrained the way in which they could enrol. Clarkson & Henry (1973) argued that the development of new predators might be an indirect cause of the increasing complexity of the coaptative structures in some of the Ordovician trilobites. However, Hammann (1985) stated that in some cases the environment played a more important role to address morphological adaptations in the head and pygidium for enrolment. Our results show that trilobites with cylindrical enrolment types (mostly odontopleurids) represent a very small portion of the sample (5%) but had a very high cephalic morphological disparity. This is suggesting an evolutionary trade-off between a relatively simple way to enclose their bodies (without the need for interlocking devices) and a high morphological disparity in the head, that may have helped the group occupy a number of ecological niches in shallow marine deposits (Hammann 1985; Mergl 2014; Esteve & Yuan 2017). Stigall *et al.* (2019) suggested that the term GOBE should be used specifically to identify a Middle Ordovician phenomenon, whereas the broader term 'Ordovician Radiation' reflect the longer, cumulative increases over the entire Ordovician Period. Our results agree with Stigall *et al.* (2019) given that the trilobites using established enrolment styles with new adaptations to improve enrolment appear throughout the Ordovician Radiations while the new enrolment types (i.e. inverted and discoidal) appeared in the Middle Ordovician, corresponding with the GOBE. In addition, although

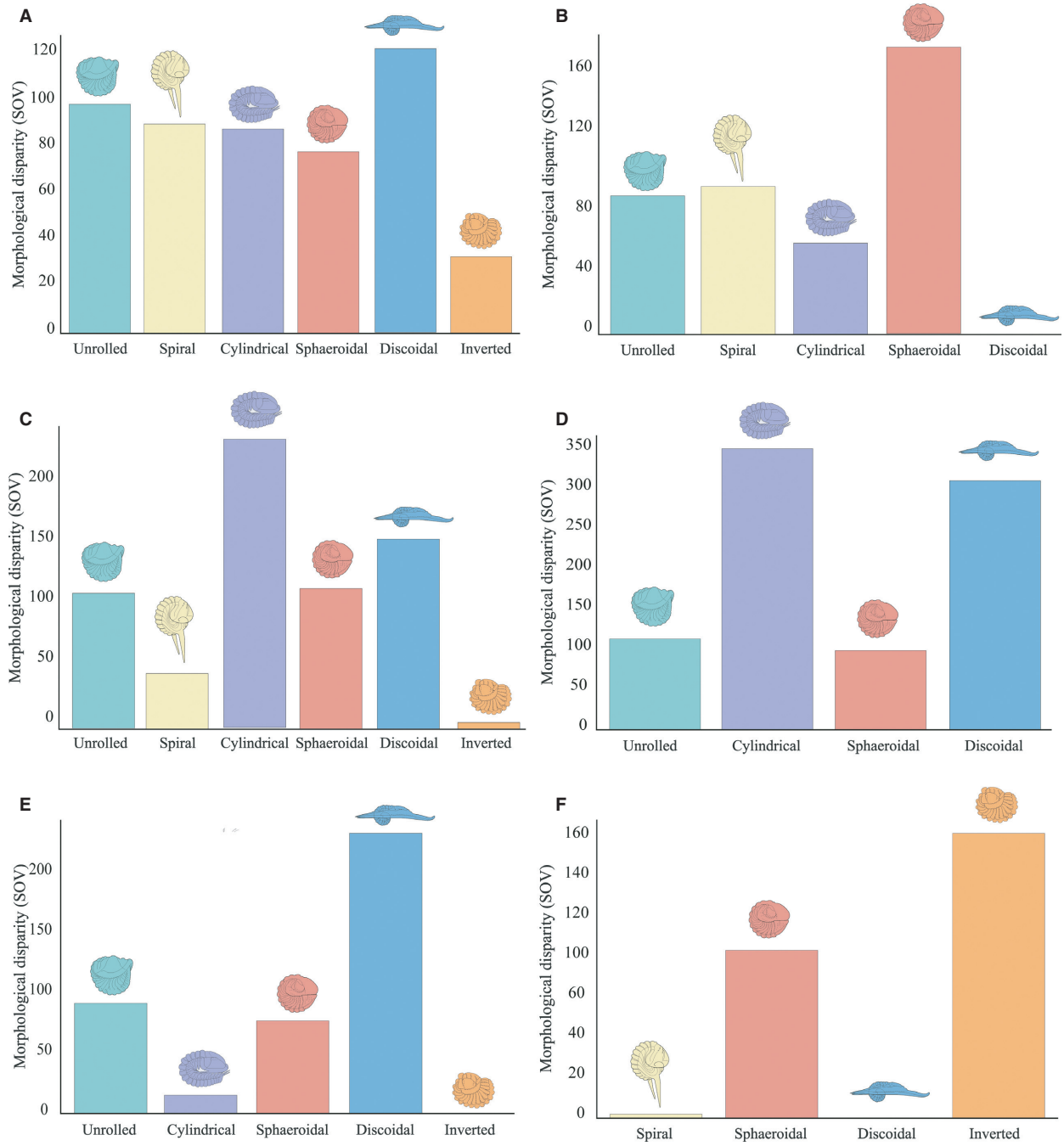


Fig. 4. Patterns of morphological disparity (sum of variances, SOV) for the six enrolment types. Distribution of variance in the enrolment types through the Palaeozoic (A), and for each period Cambrian (B), Ordovician (C), Silurian (D), Devonian (F) and Carboniferous-Permian (G).

the discoidal enrolment has been recorded in Cambrian trilobites, it was not until the Middle Ordovician when this enrolment type become more common.

In general, throughout the Palaeozoic morphospace and disparity decreases but the cephalic outline disparity of unrolled, cylindrical and discoidal enrolment types increases. We can observe how effectively the morphological diversity and disparity have evolved in a non-uniform fashion, accumulating greater disparity in the Cambrian and Ordovician where the origin of the many taxonomic groups is found. Foote (1991, 1993) found higher rates of disparity and higher morphospace occupation in the Ordovician and lower rates and morphospace occupation in the Cambrian. However,

Foote (1991, 1993) only analysed the cranidial shape including the information on the course of the dorsal facial suture (except in those forms with a marginal suture); nevertheless, our analysis here also included the outline of the free checks. The course of the dorsal facial sutures varied relatively little among most Cambrian trilobites, but the outline of the free checks seems to be more variable than in Ordovician trilobites. Adaptations associated with different enrolment types that appear in the middle Cambrian may be the reason behind this unusually high disparity. However, the morphological and disparity patterns in the middle and upper Palaeozoic observed in our analysis agree with the analysis of Foote (1991, 1993). Therefore, this non-uniform pattern reveals large changes in evolutionary rates that correspond to the generation of new morphological forms that emerged from the radiation of clades, first in the Cambrian but also during the Devonian and the Permian. The morpho-functional novelties (i.e. enrolment styles) coincide with an increase of morphospace observed during these radiations (see below). However, not all enrolment types account for the same degree of morphological disparity. Inverted enrolment type shows the lowest morphological disparity among all enrolment types, while discoidal enrolment type shows the highest rate of morphological disparity. The inverted enrolment style requires a very specific modification in the cephalic shield to fit the pleural and pygidial tips into the anterior and lateral border (see Henry & Clarkson 1974; Esteve *et al.* 2018). This prevents a high disparity in the cephalic shape of trilobites with inverted enrolment type. On the contrary, trilobites with a discoidal enrolment type bear a wide brim and the pygidium and pleural tips fit under a narrow geniculated groove below a girder subparallel to the brim. Thus, this morphology with a wide brim is recorded in a number of different genera belonging to harpids, asaphids (e.g. trinucleids) but also in one 'ptychopariid' trilobite (Bergström 1973; Ebach & McNamara 2002; Esteve *et al.* 2012). Consequently, the morphological disparity of trilobites with the discoidal enrolment type is very high. Trilobites with an inverted enrolment type were very unsuccessful (in terms of taxonomic diversity), with only a small group of phacopids during the Ordovician and proetids during the Permian possessing this enrolment type. This is another example of evolutionary trade-off between the high morphological disparity of the head and the possibility to achieve an efficient enrolment. Morpho-functional and disparity variations in our sample are directly correlated with periods of marine biodiversity radiation and extinction in the Palaeozoic. Evolution of new enrolment types during the Ordovician may have been driven by pre-Tremadocian extinctions (Saltzman *et al.* 2015; Edwards 2019; Stigall *et al.* 2019). Trilobites were affected by the end-Ordovician mass extinction in the late Katian to the Hirnantian (Fan *et al.* 2020); this is reflected in the decrease of morphospace occupation and lower disparity in the Silurian. Trilobite morphological disparity almost recovered to Ordovician levels in the Devonian where a high diversity of enrolment types existed (assisted by the high diversity of lower latitude faunas from Morocco, not illustrated here). Nevertheless, after the late Devonian, Frasnian–Famennian extinction (McLaren 1982; Feist & Clarkson 1989; Feist 1991; Alroy *et al.* 2001; Fan *et al.* 2020) the morphological diversity of cephalic shape decreases dramatically in the Carboniferous, with very few enrolment styles, some of which are represented by only one or two taxa. Fan *et al.* (2020) recognized a new biodiversification event, the Carboniferous-Permian Biodiversification Event, comparable to the GOBE on the basis of both the increase of species richness and species/genus ratio. Only four proetacean families survived the late Devonian extinction (Owens 1986) but our results agree with Fan *et al.* (2020) given the increase observed in the morphospace in the Permian together with an increase in morphological disparity. Lerosey-Aubril & Feist (2012) reported a high diversity of proetids during the late Palaeozoic and based on this argued that the extinction of trilobites was not inevitable but rather the result of major environmental changes that affected the biosphere during the Late Palaeozoic. Our results agree with Lerosey-Aubril & Feist (2012), and proetacean trilobites were a conservative group that kept a very simple morphological bauplan since Cambrian times. However, during the Carboniferous-Permian Biodiversification Event, underwent a burst of homeomorphic variation which coincide with an increase in morphological disparity. Consequently, proetacean trilobites occupied those empty niches, formerly occupied by morphologically different groups, such as the phacopid trilobites, that were made available after the Devonian mass extinction events, developing new enrolment types (e.g. inverted) and subsequently displayed a clear rebound before the late Permian extinction.

Finally, the sphaeroidal enrolment type is the only type that appears later in the Palaeozoic. Morphological disparity in the head of trilobites using sphaeroidal enrolment type suffered a small decrease in the Silurian, after the end-Ordovician mass extinction (Fan *et al.* 2020) but recovered in the Devonian and once more after the Carboniferous-Permian Biodiversification Event. Compared to trilobites possessing other enrolment types, those using sphaeroidal enrolment type show the same rate of disparity in the head

after the Cambrian. This suggests that some form of constraint in the cephalic shape may have been present to achieve sphaeroidal enrolment type. Sphaeroidal enrolment type is also the enrolment type used by a greater range of trilobite orders. Given the high number of orders with this enrolment type, we can conclude that sphaeroidal enrolment type was a very efficient way to protect the ventral side of their bodies and arose independently among different trilobite groups.

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Supporting Information

Additional supporting information may be found online in the Supporting Information section at the end of the article.

Appendix S1. Trilobite data base.